Flyweight

Share glyph styles across a million characters

Season 2 · GoF · Beginner

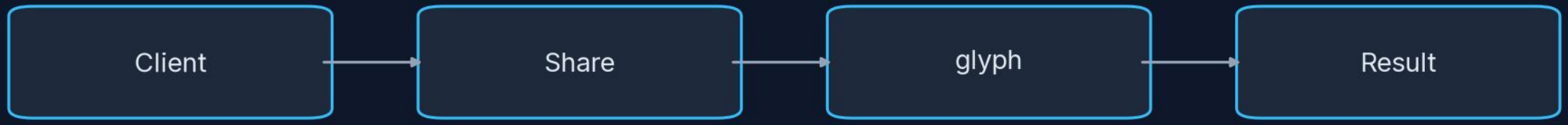
Today's Hook

- Problem: Share glyph styles across a million characters
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to Flyweight with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Share glyph styles across a million characters
- Constraints: solve «Share glyph styles across a million characters» with Flyweight — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Share glyph styles across a million characters
- Success looks like: stylesCreated=1000

Structure



Flyweight: Share glyph styles across a million characters

Real-World Scenario

- Running example: Share glyph styles across a million characters
- Key types: GlyphStyle, GlyphFactory, Glyph
- Happy-path stdout cue: stylesCreated=1000
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

Naive Code

```java
package com.buildwithwaqar.patterns.gof.structural.flyweight.problem;

import java.util.*;

public final class NaiveGlyphs {
  public static int render(int n){
    int styles=0; for(int i=0;i<n;i++) styles++; // new style each glyph
    return styles;
  }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Share glyph styles across a million characters
- Without Flyweight, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.gof.structural.flyweight.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;

final class GlyphStyle {
  private final String font; private final int size;
  GlyphStyle(String font, int size){ this.font=font; this.size=size; }
  String key(){ return font+":"+size; }
}
final class GlyphFactory {
  private final Map<String, GlyphStyle> cache = new HashMap<>();
  GlyphStyle style(String font, int size){
    return cache.computeIfAbsent(font+":"+size, k -> new GlyphStyle(font, size));
  }
  int cached(){ return cache.size(); }
}
final class Glyph {
  private final char c; private final GlyphStyle style;
  Glyph(char c, GlyphStyle style){ this.c=c; this.style=style; }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.gof.structural.flyweight.problem;

import java.util.*;
import java.util.concurrent.*;

public final class ProblemDemo {
  public static void main(String[] args) {
    System.out.println("=== Flyweight: share glyph styles ===");
    System.out.println("--- naive ---");
System.out.println("stylesCreated="+NaiveGlyphs.render(1000));
System.out.println("--- flyweight ---");
var f=new GlyphFactory();
for (int i=0;i<1000;i++) new Glyph((char)('a'+(i%26)), f.style("Inter", 12));
System.out.println("cachedStyles="+f.cached());
  }
}
```

Walkthrough Output

```
=== Flyweight: share glyph styles ===
--- naive ---
stylesCreated=1000
--- flyweight ---
cachedStyles=1
```

Step-by-Step Trace

- Match stdout to stages — look for: stylesCreated=1000
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp ... 13-flyweight ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
// Java 25 direction for Flyweight
// Prefer records + sealed where they shrink the demo.
// Problem: Share glyph styles across a million characters
```

Trade-offs

- Use Flyweight when «Share glyph styles across a million characters» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Flyweight in product language.

Related Patterns

- Name the pattern usually confused with Flyweight and separate the forces.
- Sketch the roles for «Share glyph styles across a million characters» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Share glyph styles across a million characters
- Naive pain → Flyweight boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.